Practice workbook · Embedded systems: small computers inside devices

What an embedded system is, its characteristics, examples inside everyday devices, the sensors and other hardware it controls, and why it suits a dedicated task.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-09

1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a.ii only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

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(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

1. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

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P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a.ii · 2 marks

2019 JUN · Paper 01

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3. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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4. Question 5 · part c.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.i only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

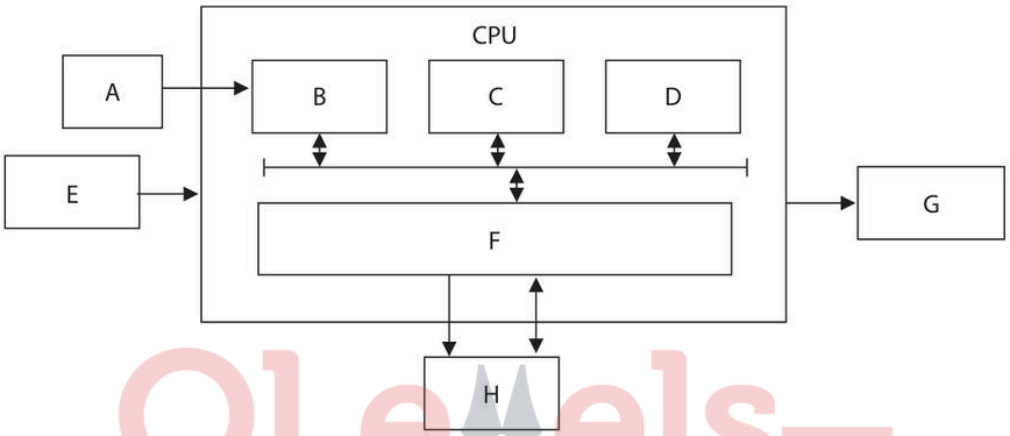


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

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4. Question 5 · part c.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.i only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

☒ **A** Memory delete signal

☒ **B** Memory read signal

☒ **C** Memory store signal

☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1 .....

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4. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

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(ii) There are many factors to consider when choosing a CPU for an embedded system.

For example, the clock speed of the CPU in a TV remote control can be low because it only has to be fast enough to create a low frequency output signal for the TV.

Explain **one other** factor that should be considered when choosing a CPU for an embedded system within a TV remote control.

(2)

(Total for Question 5 = 13 marks)

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Working and answer

5. Question 5 · part c.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

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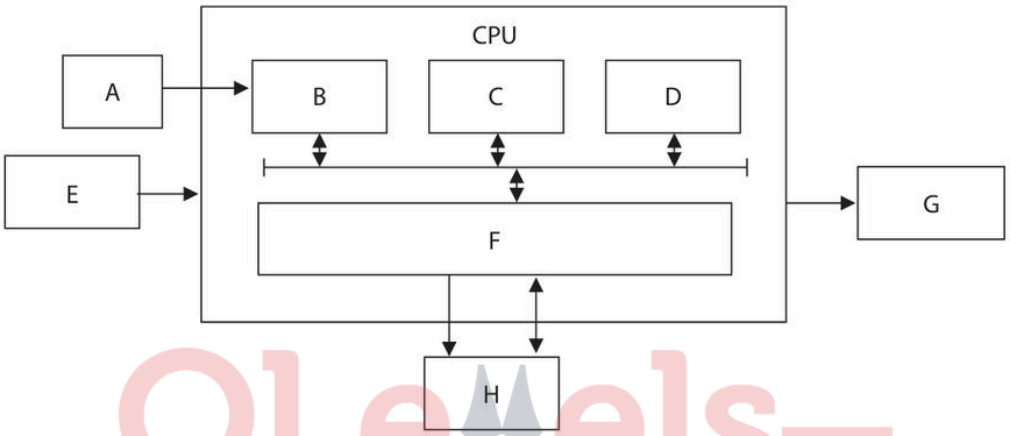


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Working and answer

6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

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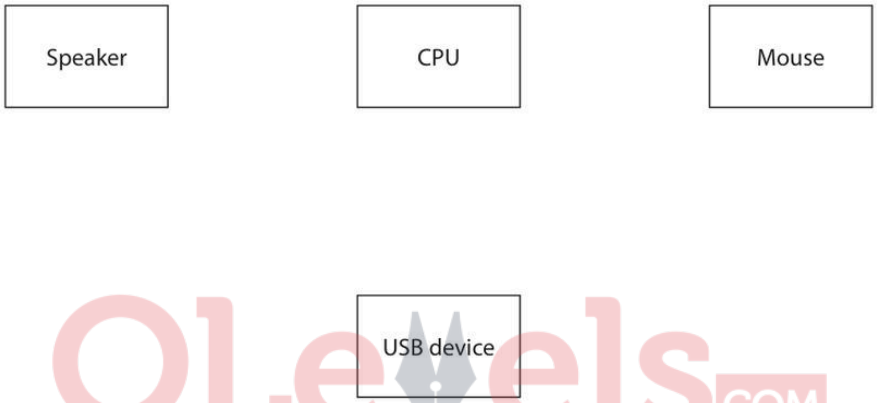
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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P 7 2 9 3 7 A 0 1 7 2 4

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Turn over ▶

6. Question 5 · part c · 2 marks · Working and answer

Working and answer

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7. Question 5 · part a.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.i only.

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P 7 8 6 3 7 A 0 1 3 2 0

13

Turn over ▶

7. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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